the calibrated quantification of uncertainty that GPs offer, which we will see in the next section of this chapter.

2.3 Going from a finite to an infinite Gaussian

We are now ready to discuss what a Gaussian process is in this section. In the same manner as when we have three variables A, B, and C or twenty in the previous section, let's say we now have an infinite number of variables, all belonging to a multivariate Gaussian. This *infinite-dimensional* Gaussian is called a Gaussian process.

Imagine predicting housing prices across a very large, densely populated area. The scale of the entire area is so large that if we move away from a house by a very small amount, we will arrive at a different house. Given the high density of the variables (houses) in this Gaussian, we can treat this whole area as having infinitely many houses; that is, the Gaussian distribution has infinitely many variables.

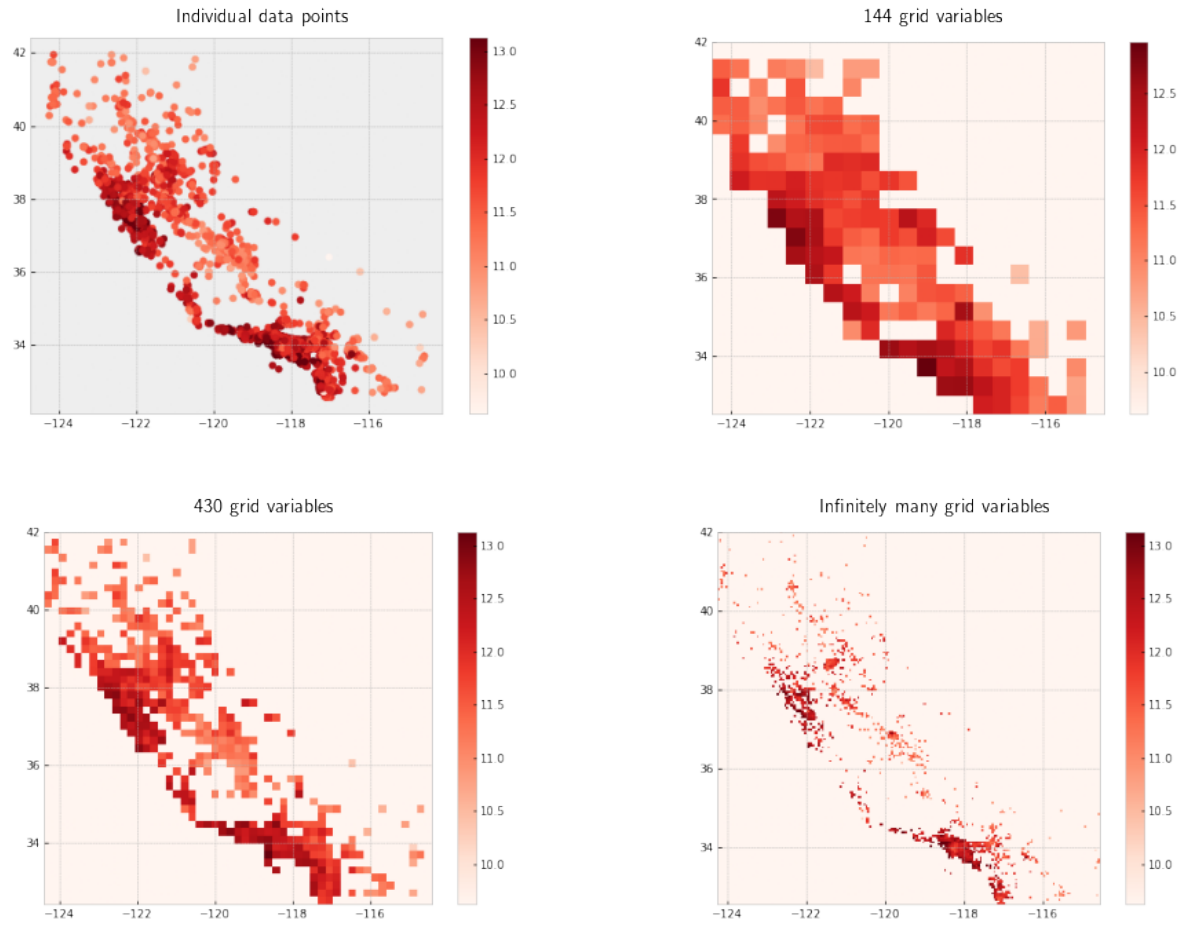


Figure 2.12 Modeling housing prices in California, using different numbers of variables. The more variables we have, the smoother our model becomes, and the closer we approach an infinite-dimensional model.

This is illustrated in Figure 2.12, using a data set containing 5000 house prices in California. In

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